Graphical Abstract

Cross-Modal Generative Alignment for Document, Chart, and Image Reasoning with Explanation-Aware Vision–Language Models

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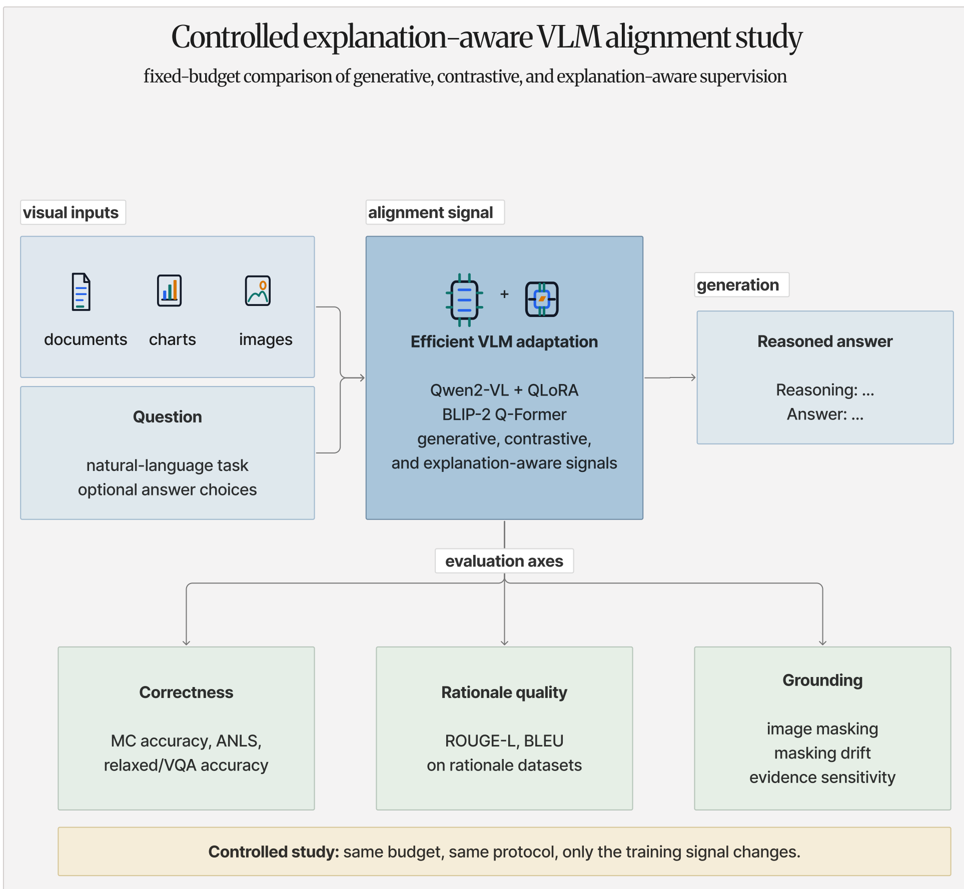

Highlights

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- The study runs zero-shot, answer-only, rationale-generative, explanation-aware, and BLIP-2 contrastive-enhanced controls under the same single-GPU environment.
- In rationale-bearing datasets, the explanation-aware training does help, but it's not the best; answer-only controls are strong.
- BLIP-2 contrastive-enhanced fine-tuning yields only a small gain in ScienceQA with BLIP-2 over generative fine-tuning, and the retrieval diagnostic are very close to frozen and random controls.
- Evidence-masking drift shows evidence sensitivity of the model, so the answer gains, rationale quality, and grounding claims are kept separate.

Cross-Modal Generative Alignment for Document, Chart, and Image Reasoning with Explanation-Aware Vision–Language Models

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Abstract

A vision-language model (VLM) is able to read a chart, document, scanned form, or natural image and answer questions about it. But how do we know if the answer is actually derived from the image itself, rather than being a lucky guess on a question based on the model's remembered language patterns? The correctness of the answer itself is not enough if the reasoning for that answer is not grounded in the visual evidence of the image. Charts and documents are a good example of this problem, because to answer the question correctly, the model has to actually ground its reasoning on specific parts of the image, making it easier to spot when the model is looking at the wrong thing. Although recent work has made progress on this issues, there was no isolation of the training objective as the main variable, with all other factors such as the model, the data, and the budget held constant. Contrastive, generative, and explanation-based methods have all been proposed, but they have not been directly compared under the same conditions, so it is not clear what the objective itself contributed to the final performance. In this paper, we keep the pipeline fixed and change only the training signal, so we can see what the objective does to answer accuracy, explanation quality, and evidence sensitivity when the model, the data, and the compute budget are all held steady. We compare five settings head to head rather than treating them as unrelated systems: BLIP-2 generative fine-tuning, BLIP-2 contrastive-enhanced fine-tuning, Qwen2-VL under answer-only, rationale-generative, and explanation-aware supervision. All the runs use the same resource-constrained single-seed protocol, and the same evaluation code as much as possible, across ScienceQA, A-OKVQA, ChartQA, DocVQA, and a VQAv2 subset. The picture that emerges is mixed. On ScienceQA, Qwen2-VL-2B reaches 0.800 accuracy with answer-only supervision, 0.780 in both rationale-generative and length-aware explanation control settings, and 0.765 with fixed-alpha explanation-aware training. On A-OKVQA, the same model reaches 0.830 with answer-only supervision, 0.790 in both rationale-generative and length-aware explanation control training, and 0.795 with fixed-alpha explanation-aware training. ChartQA, DocVQA, and the VQAv2 subset have no gold rationales, so those runs fall back to answer-only controls; they lift the native headline metrics, but they do not test rationale supervision at all. BLIP-2 barely moves: the contrastive objective gives a small gain compared to the generative one on ScienceQA 0.480 vs 0.475, but the retrieval diagnostic stays near frozen and random controls in both cases. Scale is the exception. With a fixed explanation-aware objective on ScienceQA, Qwen2-VL-7B reaches 0.885 while the 2B version reaches 0.765. The masking analysis tells us whether the model relies on the image, not whether its explanation is faithful. Overall, the results do not show a clear win for explanation-aware training. The rationale supervision can help on a single-GPU budget, but the gain is easy to mis-credit to scale or tuning. Hence, it only stands up when answer-only controls, objective balance, model scale, transfer and evidence sensitivity are each reported separately.

Keywords: vision–language models, cross-modal alignment, explanation-aware training, chart and document understanding, faithfulness, parameter-efficient fine-tuning

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1. Introduction

Vision–language models (VLMs) are becoming a default tool for answering questions based on the content of images, documents, and charts. The process looks simple: you give it an image and a question, and it will respond with an answer the image actually supports. What is considered as evidence depends entirely on the picture. In a document it might be sitting in a line of a body text, in a table, or in a stamped region of the scanned page. In a chart it might be in an axis label, the relative height of the bars, a colour, or a trend that is not written down explicitly in words. In science and commonsense visual question answering (VQA), it is often not in the image at all, forcing the model to deduce the answer from the background knowledge it has already learned. Across every one of these, a correct-looking answer is not enough if the reasoning for that answer is not actually based on what is visible in the image. The VLMs are very good at generating the right-looking answers while looking at the wrong thing. That is why in this paper we do not rely on the answer alone, but also check the quality of the explanation and the sensitivity of the answer to visual evidence.

The issue behind all of this is a cross-modal alignment problem: how do we connect the visual signal to the language model in a way that actually makes the two talk to each other? It turns out that the quality of that wiring depends on many factors, such as the visual resolution, the data mixture, the connector design, the instruction tuning, and the adaptation method [1, 2, 3]. Open model families are very helpful, because backbones such as Qwen2-VL, BLIP-2, and LLaVA-OneVision are strong enough out of the box, while still being adaptable on an academic budget [4, 5, 6]. At the same time, document and chart studies show that general visual instruction following is not enough when it comes to text-heavy and structured images [7, 8, 9, 10].

Furthermore, there is also a budget reality to consider. Most students and small labs cannot afford full fine-tuning of a large multimodal model. Methods like DoRA (weight-decomposed low-rank adaptation) make it possible to fine-tune the model by updating only a small portion of the parameters, while keeping the base model frozen or quantised [11]. Hence, in this study, we take the same approach using low-rank adaptation (LoRA) and its quantised variant (QLoRA) instead of full-model fine-tuning [12, 13]. The main goal is not scoring the highest on a leaderboard, but rather to check the model's performance when only the alignment objective changes while keeping all other factors constant.

1.1. Gap Analysis

The gap is isolation. Most papers change the model, the data, the training size, and the objective all at once. Once the final result is reported, it is hard to tell how much the objective itself contributed. Maybe the objective helped, but it might also be the case that it was simply a bigger model, cleaner data, or more tuning and not necessarily the objective itself.

Moreover, each objective has its own blind spot. Contrastive learning is good at matching an image to a caption, but it does not teach the model to generate an explanation or an answer from the image. Generative learning is good at producing an answer, but it does not necessarily mean that the model is basing its answer on the correct parts of the image. Explanation-aware training teaches the model to actually write reasoning to come to the answer, but the model could have learned to write a convincing explanation after the fact, which might not actually be the cause of the answer.

Hallucination and faithfulness evaluation has mostly been studied on natural images, where the common test is whether the model names objects that are not there. Charts and documents have recently come into focus too, with benchmarks for chart hallucination, chart grounding, and document-answer groundedness emerging [14, 15, 16]. They are still comparatively underexplored, even though their evidence is more structured and easier to verify. That is exactly what makes them a good test case for grounding.

These gaps are not closed by a new model or benchmark. What is actually missing is a controlled study that holds the model, the data, the budget, and the evaluation procedure constant while changing only the alignment objective, to test not only the answer quality but also the explanation quality and the evidence sensitivity of the model under each objective.

1.2. Research Questions

The main question is to investigate how the resource-constrained vision–language model should be aligned to achieve both accurate and faithful visual reasoning. The following research questions break that down into more measurable parts.

1. RQ1: How have vision–language alignment methods evolved for grounded visual reasoning and explanation generation?
2. RQ2: How does BLIP-2 generative fine-tuning compare with BLIP-2 contrastive-enhanced fine-tuning on visual reasoning?
3. RQ3: Does explanation-aware fine-tuning improve answer accuracy and explanation quality on datasets with gold rationales?
4. RQ4: How does the same fine-tuning framework perform across document, chart, science, commonsense, and natural-image domains?
5. RQ5: How sensitive are the model's answers to visual evidence under image-masking perturbations?
6. RQ6: How does the selected explanation-aware QLoRA setup scale from Qwen2-VL-2B to Qwen2-VL-7B?
7. RQ7: How well do fine-tuned adapters transfer across visual domains and prompt formats?

1.3. Problem Statement

The problem is cross-modal alignment in a resource-constrained environment. Given the same pretrained vision–language model, we analyse the effect of different training objectives on answer accuracy, explanation quality, and evidence sensitivity. The model, the data format, the optimiser, and the evaluation process are kept constant wherever possible, leaving only the training signal as the variable. The BLIP-2 pair reuses one Q-Former training setup. The Qwen2-VL generative and explanation-aware runs reuse one QLoRA setup. The goal is not to find the best scoring setup, but to observe whether the chosen objective leads the model in a direction that matters for a more trustworthy reasoning, and to check whether the answer quality, the explanation quality, and the evidence sensitivity move together or separately under each objective.

1.4. Novelty of this study

The contribution is the controlled comparison, not a new architecture. Instead of introducing a new architecture, we compare existing ones under different training objectives. We consider the faithfulness of the model's reasoning as a first-class result, rather than concentrating solely on answer accuracy. The paper contributes the following:

- A comparison of zero-shot, answer-only, BLIP-2 generative, BLIP-2 contrastive-enhanced, Qwen2-VL rationale-generative, and Qwen2-VL explanation-aware alignment under the same data and optimisation settings.

- Evaluation along three separate axes, namely answer quality, explanation quality, and evidence sensitivity, across document, chart, science, and commonsense tasks.

- For multiple-choice reasoning, a generate-then-score protocol is used, so the model is evaluated only after it has produced its own reasoning trace.

- A direct test of how QLoRA behaves with scale, instead of assuming that full fine-tuning or very large models are the only things worth studying.

1.5. Why this study matters

The usefulness of this study is in the constraints. It runs on a resource-constrained setup that is actually accessible to students and small labs. It also avoids concentrating solely on a single aspect of performance, and instead keeps answer accuracy, explanation quality, and evidence sensitivity as separate readings. On documents and charts especially, that separation is particularly important, because the evidence should trace back to a visible region. The whole pipeline is reusable: every dataset is transformed into one shared format, and every experiment is declared in a configuration file, so the code can be used for future comparisons of new objectives, new models, or new datasets, without worrying about hidden differences between runs. And the results are honest rather than overselling: the rationale supervision does help on the science and commonsense runs, but the answer-only controls stay strong, and the BLIP-2 contrastive objective provides only a small answer gain on top of weak retrieval evidence.

2. Literature Review

The review covers relatively latest works in the areas of vision–language model design, text-rich and document-heavy images, charts and structured visual reasoning, faithfulness and critique, and efficient adaptation. We group it by the parts that actually matter here the most: model design, text-rich images, chart reasoning, faithfulness, datasets, metrics, and low-budget adaptation.

2.1. Recent vision–language model design

Recent vision–language models are not built around a single alignment trick. Their performance is a result of many choices working together, such as the backbone architecture, the visual tokenisation method, the data mixture, the connector design, and the instruction tuning. Since all those factors are usually changed together, it is hard to tell which part actually caused an improvement in the final performance.

This is exactly the issues raised by recent ablation studies, which demonstrate that these factors are still poorly separated, and that it is vital to conduct controlled comparisons [1]. MM1.5 also makes the same argument, showing that optical character recognition (OCR) data, synthetic captions, instruction mixtures, and specialised variants can each weigh as much as raw scale in the final performance of a vision–language model [2]. Molmo and

PixMo are useful here because they provide open weights and open data that let us actually read the recipe and understand what is going on under the hood [3]. LLaVA-OneVision shows that a single family of models can be used across image and video when the data mixture is chosen with care [6].

For this study, the conclusion is simple. If the model, the data, and the budget are all changing together, then the contribution of the objective itself is hard to tell. That is why we need a controlled comparison, where the model, the data, and the budget are held constant while only the training signal changes.

Open model families are also important, because they enable us to actually run the controlled comparison on a practical budget. Qwen2-VL is designed to work with dynamic image resolution and text-heavy visual inputs such as documents, charts, and diagrams [4]. BLIP-2's Q-Former lets us test an image–answer contrastive loss while keeping the larger vision encoder and language model frozen [5]. We use these open models because they make controlled, parameter-efficient experiments feasible under limited resources.

2.2. Text-rich and document-heavy images

Documents are difficult for a specific reason. The answer in this case is not in the whole image, but rather in a small part of it, such as a line of text, a table cell, or a layout region. Hence, the task is not just an image recognition problem. The model has to read the document and find the evidence in the right place, which is a different skill than naming objects in a natural image.

Recent work shows that text-rich image reasoning is moving toward multimodal large language models, especially for OCR-heavy and layout-heavy inputs [7]. DocLLM is a good example of this trend. It encodes the document layout directly into the generative document-understanding problem, which is a step toward more faithful reasoning on documents [8]. OmniDocBench focuses on the coverage angle instead, scoring diverse PDF parsing cases with detailed annotations, which is a step toward more structured evaluation of document understanding [17]. In this study, we use DocVQA as a test case for document understanding, because it is a well-known benchmark where the answer is usually located on scanned text, layout, and local visual context [18].

Our goal is not to build a new document model. We want to see how much the training objective itself contributes to the model's ability to read documents and answer questions about them.

2.3. Charts and structured visual reasoning

Charts create a different kind of problem. In this case, the model has to read the labels, understand the axes and tick marks, and determine the relationships between the visual elements, such as bars, lines, or points. ChartInstruct and ChartGemma demonstrate that instruction tuning on chart images can help in both chart understanding and reasoning [9, 10]. ChartMimic shows the same point by training a model to generate code for that chart, which makes the model understand the chart in a more structured way [19]. ChartMuseum shows that current large visual–language models still perform far poorer than humans on chart reasoning [20]. MathVista, MMMU-Pro, and MMReason extend this concern to mathematical and multi-step reasoning, which is often required for chart understanding [21, 22, 23].

This is why we use ChartQA, ScienceQA, and A-OKVQA in the same study [24, 25, 26]. Together, they cover structured chart reasoning, science reasoning, and knowledge-heavy visual reasoning.

2.4. Faithfulness, critique, and efficient adaptation

Explanation-aware training has a subtle problem. The model might learn to write an explanation that looks convincing, but that is not actually based on the image and is not the real cause leading to the answer. Recent studies approach this problem from different angles. MMMU tests expert-level multimodal reasoning across many visual formats, to see if VLMs can do expert-style visual reasoning and not just simple image captioning [27]. MMStar checks whether a model actually needs the image to answer the question or whether the answer can be derived from the learned knowledge alone [28]. HallusionBench focuses on cases where language hallucination and visual illusion appear together in LVLM responses [29]. Hallusination-augmented contrastive learning turns a hallucination into a contrastive signal, to teach the model to avoid it [30]. VISCO and Critic-V evaluate the model's ability to critique, correct, and detect errors in its own output [31, 32]. These works suggest that faithfulness should be measured separately from answer accuracy.

Efficient adaptation is also important, because it allows us to test the effect of the training objective without needing to fine-tune the entire model. DoRA is a good example of this approach, which demonstrates that we can actually adapt the model by updating only a small portion of the parameters [11]. Reason-RFT points to another direction that is relevant here, where post-training specifically rewards visual reasoning [33]. In this paper, we stay with supervised objective comparisons, but the concern is similar. We evaluate the model as a reasoning system, rather than only as an answer generator.

Table 1 summarises the relevant works and the gaps we address in this study. It is not meant to be a comprehensive review of the field, but rather a snapshot of the recent trends and how they relate to our controlled comparison.

Table 1: Representative related work and how it connects to the gap studied here. A check mark means the line of work directly covers the property, a cross means it is not part of the main scope, and a tilde means partial coverage.

Work	Main focus	Generation	Structured VQA	Rationales	Efficient FT	Connection to this study
Recent VLM construction [1, 3, 6, 2]	Model and data recipe	✓	~	~	~	Shows that model design and training data must be controlled before we can attribute gains to the objective.
Open VLM backbones [4, 5]	Adaptable model families	✓	~	~	~	Provide practical backbones for controlled experiments under a fixed compute budget.
Text-rich document work [7, 8, 17]	Documents and PDF parsing	✓	✓	~	~	Motivates the document setting, but usually does not compare training objectives under one fixed backbone.
Chart reasoning work [9, 10, 20, 19]	Chart understanding	✓	✓	~	~	Shows that chart reasoning is still difficult and needs structured evaluation beyond ordinary image QA.
Recent reasoning benchmarks [21, 27, 28, 22, 23]	Harder multimodal evaluation	~	✓	~	✗	Motivate evaluation on visual, mathematical, expert-level, and multi-step reasoning tasks.
Faithfulness and critique [29, 30, 31, 32]	Reliability of reasoning	~	✓	✓	✗	Support the need to measure hallucination, critique, and evidence use separately from answer accuracy.
DoRA / Reason-RFT [11, 33]	Efficient and reasoning-focused adaptation	~	✗	~	✓	Motivate low-resource adaptation and explicit reasoning-oriented post-training.
This study	Controlled explanation-aware alignment	✓	✓	✓	✓	Compares objective choices directly and reports answer accuracy, rationale quality, and evidence sensitivity together.

A common pattern emerges from this review. Earlier work on multimodal alignment focused mainly on learning image–text representations, or on generating an answer from the image and the question. More recent work has started to pay attention to the faithfulness of the model's reasoning, and to the quality of the explanations it generates. However, it is still unclear how much the training objective itself contributes to the model's performance. In this study, our goal is to fill that gap by conducting a controlled comparison of different training objectives while keeping all other factors constant.

3. Methodology

The main rule in this methodology is to keep the pipeline unchanged and modify only the training objective. Each dataset is transformed into a uniform format. Then, the model is loaded, QLoRA adapters are applied if needed, the model is trained with the chosen objective, and evaluation is performed using the same generate-then-score protocol for every run. Figure 1 shows the full pipeline.

We intentionally keep the pipeline simple and controlled, because the goal is to isolate the effect of the training objective, not to find the best possible score on each dataset. Every experiment is defined through a configuration file that declares the dataset, the backbone, the objective, the adapter settings, the metrics, and the seed. The same implementation is used for baseline evaluation, fine-tuning, and final evaluation, reducing the risk of hidden differences affecting the results.

3.1. Dataset

The experiments use five public datasets. DocVQA represents document understanding, where the answer is usually based on scanned text, layout and local regions of the image [18]. ChartQA is used for chart reasoning, in which the answer might depend on labels, axes, values, or relationships between elements in the chart [24]. ScienceQA provides multimodal science questions with multiple-choice answers and explanations, which is used for rationale supervision [25]. A-OKVQA adds commonsense questions and rationales over natural images, which is also used for rationale-supervised training [26]. VQAv2 is used as a general natural-image control, where the answer is often based on object recognition and simple relationships in the image [34].

There is one important detail about the VQAv2 subset. The experiment uses the configured `Erland/VQAv2-sample` source, which has 900 training examples and 100 evaluation examples. Hence, we report it as a VQAv2 subset result, not as a full VQAv2 benchmark score.

All the datasets are transformed into a uniform format, where each example is comprised of an image, a question, an answer, and optionally a rationale. If the dataset does not have gold rationales, the rationale field is left empty, and the training falls back to answer-only supervision.

3.2. Detailed methodology

Figure 2 shows the model-level setup. All backbones are loaded in the same way, and the same training code is used for every run, so that we don't have to design a separate implementation for each model or each objective. In case of Qwen2-VL, the model is loaded in 4-bit precision and trained with rank-16 QLoRA adapters [12, 13]. For the BLIP-2 contrastive run, the vision

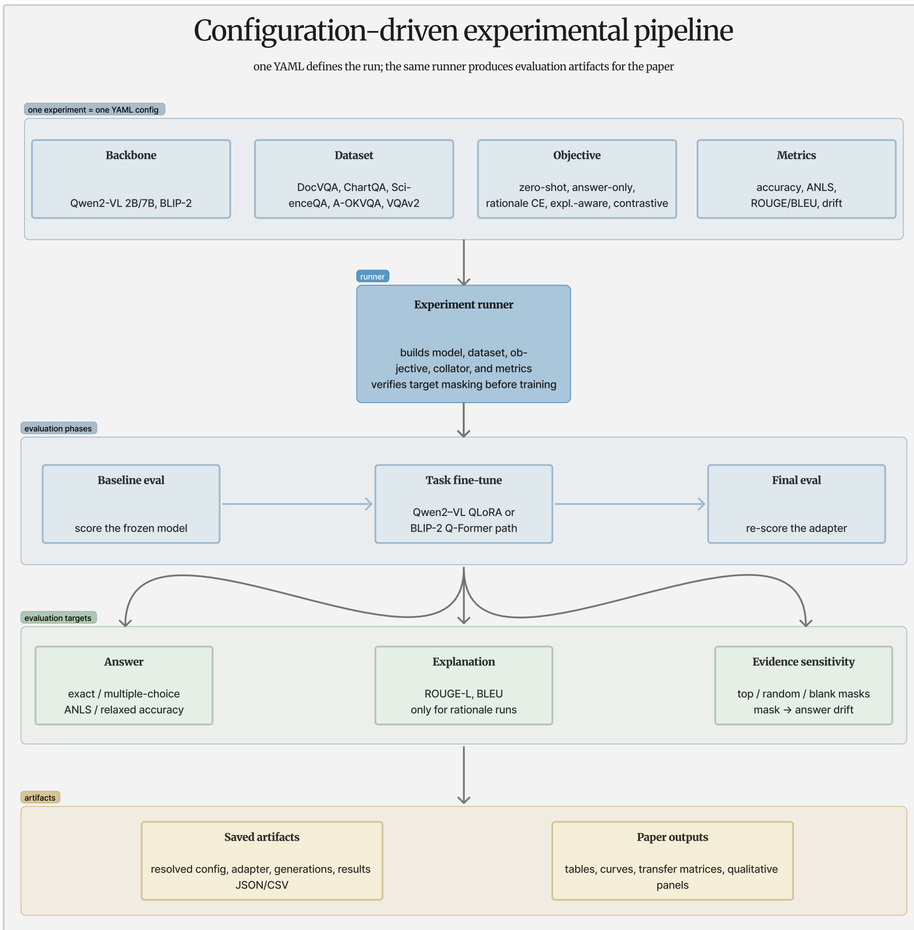

Figure 1: Overall workflow of the proposed explanation-aware alignment study. Datasets are normalised into a shared multimodal example format. Qwen2-VL runs are adapted with QLoRA, BLIP-2 runs train the Q-Former-side alignment path, and each objective is evaluated through answer, rationale, and evidence-sensitivity metrics.

encoder and the OPT language model are frozen, while the Q-Former-side components, the language projection, and a small answer-projection head are trainable.

The difference between the variants is only in the supervision signal. The baseline is evaluated before the model fine-tuning. The generative vari-

Table 2: Benchmarks used in the study. The sizes are rounded public benchmark sizes before filtering by image availability, answer type, or rationale availability. The completed experiments use capped train and evaluation splits so that every run fits the same practical compute budget.

Dataset	Domain	Approx. size	Answer type	Gold rationale	Primary metric
DocVQA [18]	Document	50K	short text span	no	ANLS
ChartQA [24]	Chart	32K	numeric / short text	no	relaxed accuracy
ScienceQA [25]	Science image	21K	multiple choice	yes	accuracy
A-OKVQA [26]	Commonsense image	25K	multiple choice / short text	yes	accuracy
VQAv2 subset [34]	Natural image	900/100 run split	open ended	no	VQA accuracy

Table 3: Input construction examples. The complete token, label, span, and collator traces are saved as pipeline artifacts; this table gives the paper-level view of what each prompt family means.

Prompt family	Raw fields	Prompt seen by model	Supervised target	Supervised spans
Answer-only	image, question, answer	Question: ... plus choices when available	Answer: <answer>.	answer tokens only
Rationale-generative	image, question, answer, gold rationale	Question: ... plus choices when available	Reasoning: <rationale> Answer: <answer>.	full target under uniform CE
Explanation-aware	image, question, answer, gold rationale	Question: ... plus choices when available	Reasoning: <rationale> Answer: <answer>.	rationale and answer spans weighted separately
Answer-only fallback	image, question, answer, no gold rationale	Question: ...	Answer: <answer>.	answer tokens only

ant is trained to produce the answer, given the image and question. The contrastive-enhanced variant adds an extra image–answer loss, which pulls the Q-Former pooled image representation closer to the correct answer and pushes it farther from the other answers in the batch. The explanation-aware variant trains the model to generate a rationale first, then the answer, and it weights the rationale and answer spans separately in the loss.

For the generative training, the prompt and targets are concatenated into one sequence, and the loss is computed only over the target tokens. The loss mask is constructed after multimodal encoding, so that the image placeholder tokens included by the processor do not shift the target span off position. This is achieved by measuring the prompt length with the same processor that encodes the full sequence, and masking out that many tokens from the start of the loss.

For contrastive objective, the same span information is used. Only the answer tokens are pooled into the InfoNCE text representation [35]. The rationale tokens still receive the generative loss, but they are not used in the contrastive loss. Equation 1 weights these two parts using α. When $\alpha = 1$, only the answer span is weighted, even if the target may still contain the rationale before the answer. That is why the true answer-only control is reported separately with a target template that contains only `Answer:`. When

$\alpha = 0$, only the explanation span is supervised. The intermediate values are used to test the interaction between answer and rationale supervision, whether they help each other or compete.

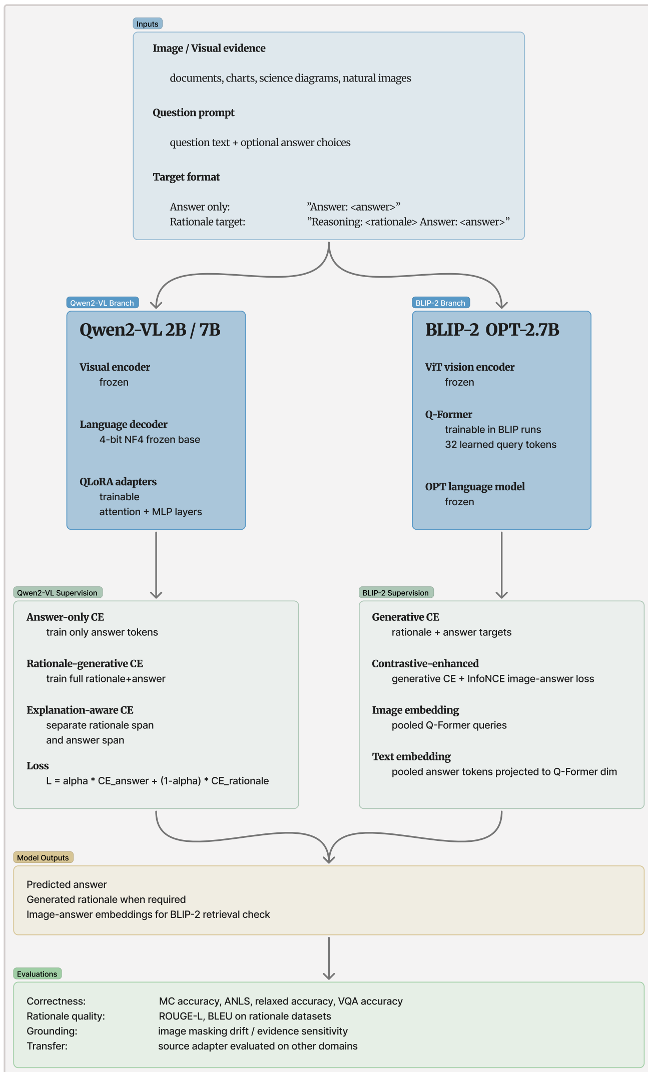

Figure 2: Model-level architecture for explanation-aware cross-modal alignment. Visual evidence and the question prompt are encoded into a multimodal token stream. The Qwen2-VL path uses trainable QLoRA adapters and separate rationale and answer supervision spans. The BLIP-2 path keeps the vision tower and OPT language model frozen and trains the Q-Former-side generative or contrastive alignment components.

$$\mathcal{L} = \alpha\, \mathcal{L}_{\text{answer}} + (1-\alpha)\, \mathcal{L}_{\text{explanation}} \tag{1}$$

Here, $\mathcal{L}_{\text{answer}}$ is the masked cross-entropy loss on the answer tokens, and $\mathcal{L}_{\text{explanation}}$ is the same loss on the explanation tokens. The coefficient α controls the relative weight of the answer and explanation supervision.

We use two ways of setting α. In fixed-alpha mode, α is chosen explicitly in the experiment configuration serving as an ablation control. In length-aware mode, the effective answer weight is computed from the number of supervised answer and explanation tokens:

$$\alpha_{\text{eff}} = \frac{\lambda\, n_{\text{answer}}}{\lambda\, n_{\text{answer}} + n_{\text{explanation}}}, \tag{2}$$

where n_{answer} and $n_{\text{explanation}}$ are the answer and explanation token counts after collation, and λ is the answer-weight multiplier. When $\lambda = 1$, the loss becomes the usual token-level cross-entropy over the full rationale-plus-answer target. Because of this, the fixed-alpha runs are treated as the ablation, while the length-aware run is used as the natural token-weighted control.

For multiple-choice datasets, inference is done in two steps. First, a short reasoning sequence is generated given the image and the question. Then, given that reasoning trace, the model scores each answer option by computing the length-normalised log-likelihood of that option as a continuation of the reasoning. Afterwards, the option with the highest score is selected as the final answer. This avoids relying on fragile string parsing or on whether the model copied an option label correctly. In case of open-ended datasets, the model just generates an answer string, which is evaluated using the dataset's native metric. When a rationale is generated, it is saved for later use in the rationale quality and evidence sensitivity evaluation.

3.3. Evaluation metrics

Evaluation is divided into three parts: answer quality, rationale quality, and evidence sensitivity. Answer quality is measured using the native metric of the benchmark, which is the primary metric reported on the leaderboard for that dataset. We use the term "headline metric" to refer to this native primary metric, not to some universal metric to be compared directly across datasets. For ScienceQA and A-OKVQA, the headline metric is multiple-choice accuracy. For DocVQA, it is average normalised Levenshtein similarity (ANLS), as per the benchmark convention [18]. For ChartQA, it is relaxed accuracy for numerical answers [24]. For the VQAv2 subset, it is VQA consensus accuracy, which is computed wherever annotator answers exist [36, 34].

For rationale quality, we use BLEU and ROUGE-L whenever gold rationales are available [37, 38]. These metrics do not measure faithfulness or correctness of the rationale, but they do provide a basic check on whether the generated explanation is lexically similar to the gold rationale, and whether it captures the same sequence of reasoning steps.

Evidence sensitivity is measured in an iterative way. The model's answer is scored on the original image, and then scored again several times by masking out different regions of the image. This is a general idea of occlusion sensitivity analysis [39], adapted to the question-answering setting. The drift in the answer score when a region is masked out indicates how much the model's answer relies on that region as evidence. That evidence-masking drift is shown in Equation 3. For qualitative inspection, we compare the regions with high evidence-masking drift to the content named in the generated rationale, to see if the model is actually naming the evidence it relies on.

$$\mathrm{drift}(r) = s(a \mid I) - s(a \mid I \setminus r) \tag{3}$$

where $s(a \mid I)$ is the length-normalised log-likelihood of the gold answer a given image I, and $I \setminus r$ is the same image with region r occluded. A large positive drift says the masked region mattered. A drift near zero says the opposite: the answer probably leaned on language priors, a shortcut, or non-visual context. The absolute scale does not travel across datasets, since answer length, prompt family, image resolution, and task format all bend the likelihood scores. The split it captures is the familiar one between a useful explanation and a faithful one. It also speaks to recent hallucination and critique work, where a model happily describes or relies on visual content that simply is not there [29, 30, 31, 32]. That is why evidence sensitivity sits next to answer accuracy here, rather than being parked as a post-hoc example.

3.4. Experimental settings

Every variant runs under the same fixed-budget protocol, and any deviation from it is written into the experiment ledger. The Qwen2-VL runs share one recipe: AdamW, a learning rate of 2×10^{-4}, cosine decay, three percent warm-up, weight decay of 0.01, bf16 compute, and an effective batch size of 32 reached through gradient accumulation. The base model loads under 4-bit NormalFloat (NF4) quantisation. Unless a result table says otherwise, the QLoRA adapters use rank 16 and scaling factor 32 [12, 13]. The workhorse backbone is Qwen2-VL-2B-Instruct, strong enough for the target tasks yet small enough to fit the budget [4]. For BLIP-2 we use Salesforce BLIP-2 OPT-2.7B, vision tower and OPT language model frozen, training only the Q-Former-side alignment components [5]. The scale ablation reuses the same QLoRA recipe on Qwen2-VL-7B. Each run logs its dataset split, filtered count, seed, objective, adapter configuration, and metrics. The pipeline also writes a split-and-leakage audit from the selected train and evaluation identifiers, wherever the dataset exposes stable IDs. For the capped datasets in the final artifact set, that overlap count comes back at zero. DocVQA is the one non-standard split: the loaded dataset only exposes validation and test, so the controlled run trains on `validation[:80%]` and evaluates on `validation[80%:]`, with native-ID overlap checking. The audit shows zero native-ID overlap, but it cannot prove document-level disjointness past the IDs the loader exposes. So we treat the DocVQA number as a capped internal adaptation result, not a public-test benchmark claim. We also attach approximate uncertainty estimates to the accuracy-style headline metrics. With most evaluation caps at 200 examples and only a few seeds in the budget, a gap like 0.790 versus 0.795 is an observed difference, nothing more. Table 4 lists the default configuration.

Table 4: Default Qwen2-VL QLoRA fine-tuning configuration. BLIP-2 contrastive runs use the separately reported frozen-vision/frozen-LLM Q-Former training setup. Any run that changes these values is reported separately in the experiment ledger and in the Results section.

Training Configuration	
Adapter	QLoRA (4-bit NF4, rank 16, scaling 32)
Optimizer	AdamW (β_1=0.9, β_2=0.999)
Learning rate	2×10^{-4} (cosine, 3% warm-up)
Weight decay	0.01
Effective batch size	32 with gradient accumulation
Precision	bf16 compute
Primary seed policy	one seed per complete configuration

3.5. Experiment ledger

Before any scores, Table 5 lays out the comparisons. The line that matters is rationale availability. ScienceQA and A-OKVQA can carry rationale-generative and explanation-aware supervision, because they ship gold rationales. ChartQA, DocVQA, and VQAv2 stay on answer-only fallback, plus the transfer and domain runs, until some future experiment adds rationale supervision of their own.

Table 5: Main experiment ledger. The full generated ledger includes exact caps, seeds, prompt variants, alpha mode, and configuration hashes.

Comparison	**Runs included**	**Use in the paper**
BLIP-2 objective pair	BLIP-2 OPT-2.7B on ScienceQA: generative and generative+contrastive.	Tests whether the auxiliary contrastive objective adds anything beyond BLIP-2 generative fine-tuning.
ScienceQA controls	Qwen2-VL-2B on ScienceQA: answer-only, rationale+answer CE, fixed-α explanation-aware sweep, and length-aware control.	Separates answer supervision, rationale supervision, and alpha weighting on a gold-rationale dataset.
A-OKVQA controls	Qwen2-VL-2B on A-OKVQA: answer-only, rationale+answer CE, fixed-α explanation-aware, and length-aware control.	Repeats the rationale-supervision comparison on a second gold-rationale domain.
Fallback domains	Qwen2-VL-2B on ChartQA, DocVQA, and VQAv2 with answer-only fallback prompts.	Reports in-domain adaptation and transfer for datasets without gold rationales; these rows are not used as explanation-aware evidence.
Scale	Qwen2-VL-2B and Qwen2-VL-7B on ScienceQA with the same fixed explanation-aware QLoRA setting.	Checks whether the selected setup changes when the backbone scale increases.
Transfer	Qwen2-VL-2B source adapters evaluated on target-domain prompts across all datasets.	Tests cross-domain behaviour separately from the official in-domain result table.
Masking	Selected Qwen2-VL runs with image-region perturbation and saved visual examples.	Measures evidence sensitivity beside answer accuracy and rationale quality.

4. Results

The results track the comparison types from the ledger, in order. The BLIP-2 objective pair comes first, testing whether the auxiliary contrastive term earns its place beyond generative fine-tuning. Then the ScienceQA and A-OKVQA runs pull answer-only, rationale-generative, and explanation-aware supervision apart on the two datasets that have gold rationales. The chart, document, and VQAv2 runs follow, showing in-domain adaptation where no gold rationales exist. After that come masking-based evidence sensitivity, the Qwen2-VL scale check, and cross-domain transfer. One reminder throughout: every score comes from the capped single-seed protocol above, so they are controlled framework results, not leaderboard numbers.

4.1. Objective-level comparison

Table 6 holds the direct objective comparisons. Take the BLIP-2 ScienceQA pair first: both trained variants clear the 0.445 frozen baseline, the contrastive-enhanced run landing at 0.480 and the generative control at 0.475. That gap is tiny, and it sits inside the approximate uncertainty intervals in Table 7. Retrieval tells the same cautious story: the contrastive-enhanced model posts recall@1 (R@1) of 0.010, R@5 of 0.020, R@10 of 0.030, and a mean reciprocal rank (MRR) of 0.0307. Qwen2-VL-2B is stronger across the board. Rationale-generative fine-tuning reaches 0.780 on ScienceQA and 0.790 on A-OKVQA. On A-OKVQA the explanation-aware run reaches 0.795 accuracy, 0.337 ROUGE-L, and 0.087 BLEU, numerically a hair above the generative control, which we read as an observed single-run difference rather than a settled win. The honest surprise is the answer-only controls: 0.800 on ScienceQA and 0.830 on A-OKVQA, both strong. The zero-shot column, meanwhile, is one shared backbone–dataset reference for orientation, not a fresh prompt-family baseline per row. Back on ScienceQA, the fixed $\alpha = 0.50$ explanation-aware run reaches 0.765. The length-aware control reaches 0.780, level with the rationale-generative control but still under the answer-only number.

Table 6: Objective-level results under the controlled protocol. The zero-shot reference is the shared pre-fine-tuning score for the same backbone and dataset group, not a separately recomputed frozen score for every prompt family. Accuracy is multiple-choice accuracy for ScienceQA and A-OKVQA.

Backbone	Dataset	Objective	Zero-shot ref.	Final	ROUGE-L	BLEU
BLIP-2 OPT-2.7B	ScienceQA	Generative	0.445	0.475	–	–
BLIP-2 OPT-2.7B	ScienceQA	Contrastive-enhanced	0.445	0.480	–	–
Qwen2-VL-2B	ScienceQA	Answer-only	0.400	**0.800**	–	–
Qwen2-VL-2B	ScienceQA	Rationale+answer CE	0.400	0.780	0.693	0.559
Qwen2-VL-2B	ScienceQA	Explanation-aware, $\alpha = 0.50$	0.400	0.765	0.643	0.493
Qwen2-VL-2B	ScienceQA	Explanation-aware, length-aware	0.400	0.780	0.696	0.563
Qwen2-VL-2B	A-OKVQA	Answer-only	0.355	**0.830**	–	–
Qwen2-VL-2B	A-OKVQA	Rationale+answer CE	0.355	0.790	0.329	0.083
Qwen2-VL-2B	A-OKVQA	Explanation-aware, $\alpha = 0.50$	0.355	0.795	0.337	0.087
Qwen2-VL-2B	A-OKVQA	Explanation-aware, length-aware	0.355	0.790	0.330	0.083

Table 7: Approximate uncertainty for selected accuracy-style headline metrics. These intervals use the generated normal-approximation standard errors for capped evaluation sets, so they are a caution about small observed differences rather than a replacement for multi-seed testing.

Run	Metric	Eval cap	Score	Approx. 95% CI
BLIP-2 generative ScienceQA	MC accuracy	200	0.475	0.406–0.544
BLIP-2 contrastive ScienceQA	MC accuracy	200	0.480	0.411–0.549
ScienceQA answer-only	MC accuracy	200	0.800	0.745–0.855
ScienceQA rationale+answer CE	MC accuracy	200	0.780	0.723–0.837
ScienceQA explanation-aware, $\alpha = 0.50$	MC accuracy	200	0.765	0.706–0.824
ScienceQA explanation-aware, length-aware	MC accuracy	200	0.780	0.723–0.837
A-OKVQA answer-only	MC accuracy	200	0.830	0.778–0.882
A-OKVQA rationale+answer CE	MC accuracy	200	0.790	0.734–0.847
A-OKVQA explanation-aware, $\alpha = 0.50$	MC accuracy	200	0.795	0.739–0.851
Qwen2-VL-7B ScienceQA, $\alpha = 0.50$	MC accuracy	200	0.885	0.841–0.929

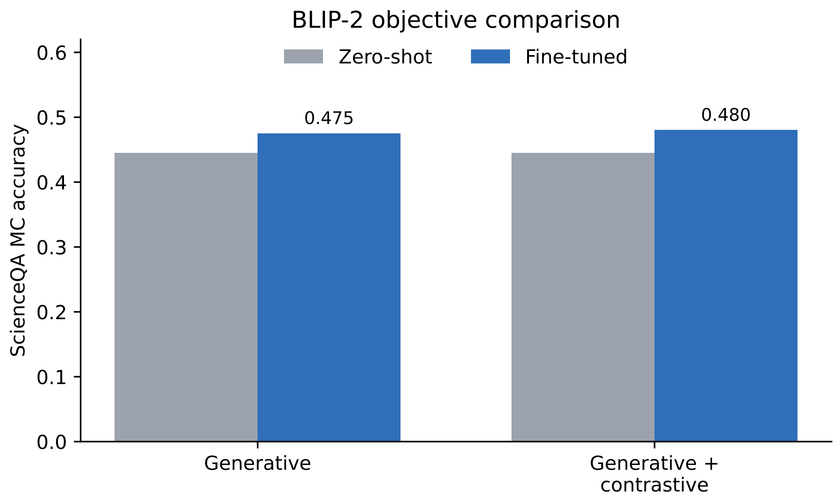

Figure 3: BLIP-2 generative and contrastive-enhanced ScienceQA results. Both runs improve over the frozen baseline, with only a small observed gain from the auxiliary contrastive term.

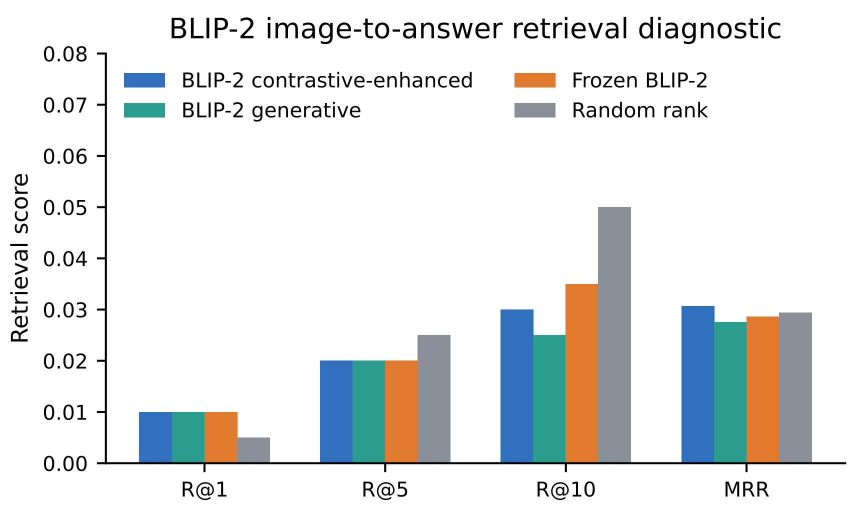

Figure 4: BLIP-2 image-to-answer retrieval diagnostic. The analysis reports frozen BLIP-2, BLIP-2 generative, BLIP-2 contrastive-enhanced, and random-rank controls so the contrastive branch is not interpreted without a reference point.

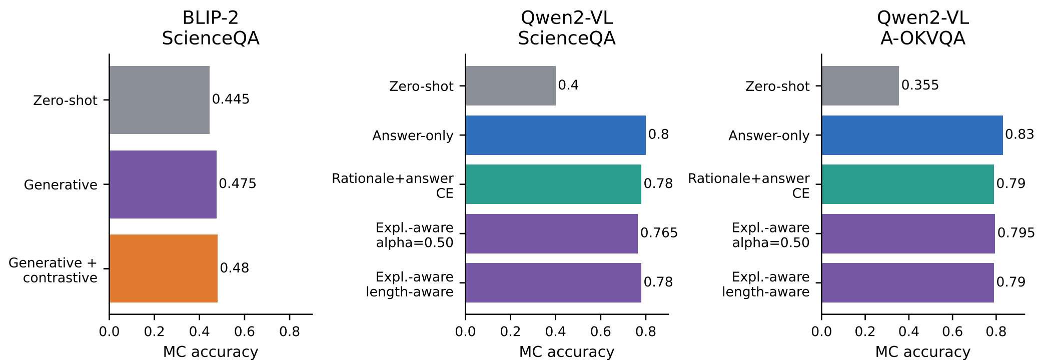

Figure 5: Controlled strategy comparison for BLIP-2 and Qwen2-VL. The figure shows zero-shot, answer-only, rationale-generative, explanation-aware, and contrastive variants where those controls are available.

4.2. Effect of explanation-aware training

Table 8 and Figure 6 trace the ScienceQA α-sweep, where α is the answer-span weight from Equation 1. Fine-tuning alone lifts accuracy from the 0.400 zero-shot reference into the 0.690–0.780 band. The sweet spot is low: $\alpha = 0.10$, at 0.780. Push α toward the answer-heavy end and accuracy slides, to 0.690 at $\alpha = 0.75$ and 0.695 at $\alpha = 1.00$. The explanation metrics move in lockstep. ROUGE-L and BLEU also peak at $\alpha = 0.10$ (0.695 and 0.566) and decay as α climbs. At $\alpha = 1.00$ only the answer span is supervised inside a rationale-shaped target, so there are no rationale metrics to report. In accuracy, the $\alpha = 0.10$ run ties the rationale-generative and length-aware controls. One trap worth flagging: the $\alpha = 1.00$ row is not the true answer-only control, because the model still sees a rationale-shaped template even though the rationale span goes unsupervised. The genuine answer-only ScienceQA control, on its own prompt family, reaches 0.800. And all of this rests on one seed, 2,000 training and 200 evaluation examples, so these are observed effects, not final statistical proof.

Table 8: Explanation-aware fixed-α sweep on ScienceQA (Qwen2-VL-2B, QLoRA, 2,000 train / 200 eval, single seed). The coefficient α weights the answer span in Equation 1: $\alpha = 1$ is answer-span-only within the rationale target and $\alpha = 0$ is explanation-span-only. The zero-shot and rationale-generative rows are references. Random-choice accuracy is 0.354. The best fixed α is shown in bold.

Variant (α)	**MC accuracy**	**ROUGE-L**	**BLEU**
Zero-shot (frozen)	0.400	–	–
Rationale+answer CE	**0.780**	0.693	0.559
$\alpha = 0.00$ (explanation-only)	0.725	0.417	0.299
$\alpha = 0.10$	**0.780**	**0.695**	**0.566**
$\alpha = 0.25$	0.775	0.686	0.548
$\alpha = 0.50$	0.765	0.643	0.493
$\alpha = 0.75$	0.690	0.604	0.443
$\alpha = 1.00$ (answer-span only)	0.695	–	–

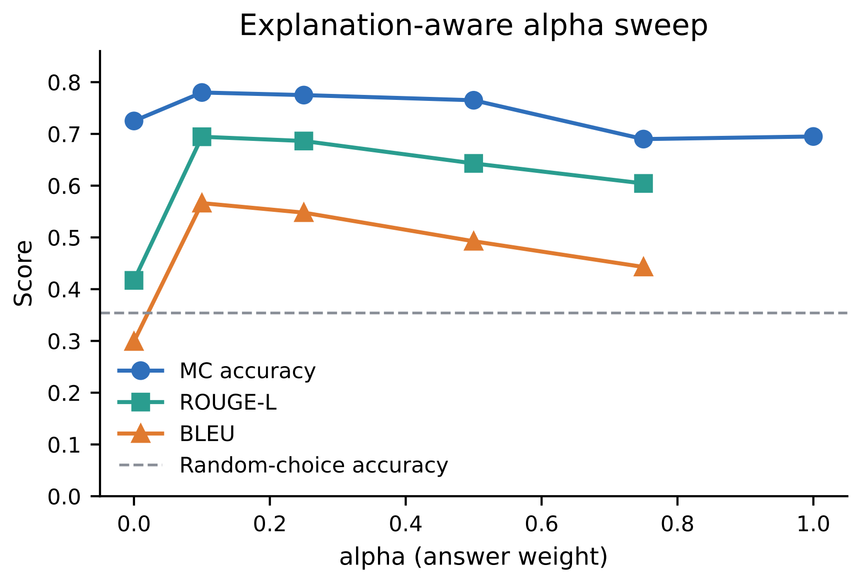

Figure 6: Fixed-alpha explanation-aware sweep on ScienceQA. Answer accuracy peaks at the low-answer-weight setting and declines toward both extremes inside the rationale-plus-answer target. Explanation quality (ROUGE-L, BLEU) peaks at the same point and falls as α increases. Series are distinguished by marker shape and line style; the dashed grey line marks random-choice accuracy. Single seed.

4.3. Domain-level behaviour

Table 9 and Figure 7 gather the selected domain runs from the final matrix. For ScienceQA the row uses the fixed $\alpha = 0.50$ explanation-aware run, so it lines up with the in-domain figure; the stronger $\alpha = 0.10$ and length-aware settings live in the sweep table above. The biggest jumps land on DocVQA, VQAv2, ChartQA, and A-OKVQA. A-OKVQA climbs from 0.355 to 0.795 accuracy against the reported zero-shot reference. The VQAv2 subset goes from 0.350 to 0.730 VQA accuracy. ChartQA reaches 0.655 relaxed accuracy, exact match 0.410. DocVQA is the headline, at 0.905 ANLS with exact match at 0.855. That number is strong for a capped internal run, but it deserves caution on three counts: the qualitative samples are short extractive answers, the setup is answer-only fallback rather than gold rationales, and the split audit cannot prove document-level disjointness past the native IDs it can see.

Table 9: Domain-level results for the completed Qwen2-VL-2B runs. Each domain uses its native headline metric: multiple-choice accuracy for ScienceQA and A-OKVQA, relaxed accuracy for ChartQA, ANLS for DocVQA, and VQA accuracy for the VQAv2 subset.

Domain	**Dataset**	**Objective**	**Zero-shot ref.**	**Final**	**Secondary metric**
Science	ScienceQA	Explanation-aware, $\alpha = 0.50$	0.400	0.765	ROUGE-L 0.643, BLEU 0.493
Commonsense	A-OKVQA	Explanation-aware, $\alpha = 0.50$	0.355	0.795	ROUGE-L 0.337, BLEU 0.087
Charts	ChartQA	Answer-only fallback	0.335	0.655	Exact match 0.410
Documents	DocVQA	Answer-only fallback	0.349	0.905	Exact match 0.855
Natural images	VQAv2 subset	Answer-only fallback	0.350	0.730	Exact match 0.350 $\rightarrow$ 0.730

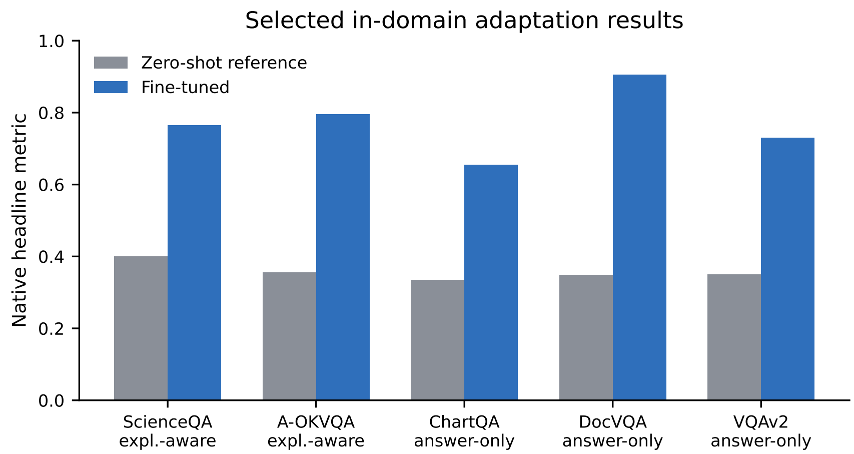

Figure 7: In-domain adaptation results for the selected completed Qwen2-VL-2B domain runs. The comparison mixes rationale-supervised ScienceQA/A-OKVQA rows with answer-only fallback rows for domains without gold rationales, so it should not be read as an explanation-aware effect across all domains.

4.4. Faithfulness and hallucination analysis

Table 10 and Figure 8 carry the evidence-masking drift check. Each number is a mean over 30 evaluation examples. The bigger and more positive it is, the more masking a region costs the gold-answer likelihood, which means the model was relying on what we hid. The drift piles up on the structured tasks: DocVQA at 0.375, ChartQA at 0.224, VQAv2 at 0.111. ScienceQA and A-OKVQA usually sit far lower, with one loud exception, the ScienceQA answer-only adapter at 0.0898. That low ScienceQA drift is not surprising; the questions come loaded with language priors and explicit answer choices, the visual evidence is diffuse, and a coarse mask can miss the single cue the model actually used. Explanation supervision does not move drift in any clean way here. At $\alpha = 0.10$ it is 0.0223, at $\alpha = 0.50$ it is 0.0197, and answer-span-only $\alpha = 1.00$ gives 0.0149, despite that run producing no supervised rationale at all. On A-OKVQA the explanation-aware run drifts a touch more than the generative control, 0.0271 against 0.0245, but the intervals around both are wide. Read plainly, the masking result backs a domain-sensitivity story, not the sweeping claim that explanation-aware training by itself buys faithfulness. We report no manual hallucination rate, because we never ran a fixed annotation protocol for hallucination categories.

Table 10: Evidence-masking drift by completed run. Higher values indicate stronger sensitivity to masked visual evidence. The intervals are approximate 95% intervals from the generated masking artifact.

Run	**Mean drift**	**Approx. 95% CI**	**Examples**
ScienceQA answer-only	0.0898	0.0197–0.1830	30
ScienceQA rationale+answer CE	0.0217	-0.0100–0.0582	30
ScienceQA $\alpha = 0.10$	0.0223	-0.0080–0.0568	30
ScienceQA $\alpha = 0.50$	0.0197	-0.0119–0.0552	30
ScienceQA answer-span-only $\alpha = 1.00$	0.0149	-0.0274–0.0638	30
A-OKVQA rationale+answer CE	0.0245	-0.0236–0.0726	30
A-OKVQA explanation-aware	0.0271	-0.0075–0.0627	30
ChartQA answer-only fallback	0.2237	0.1419–0.3140	30
DocVQA answer-only fallback	0.3749	0.2770–0.4960	30
VQAv2 answer-only fallback	0.1109	0.0274–0.2408	30
Qwen2-VL-7B ScienceQA $\alpha = 0.50$	0.0114	-0.0059–0.0260	30

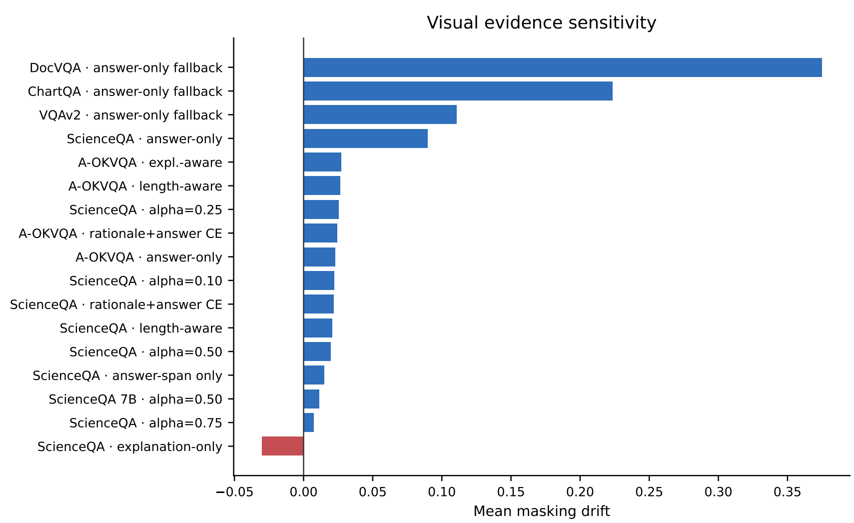

Figure 8: Mean evidence-masking drift across completed runs. Structured document and chart runs show much stronger visual sensitivity than the rationale-rich ScienceQA and A-OKVQA runs, but stronger drift does not automatically imply better answer accuracy.

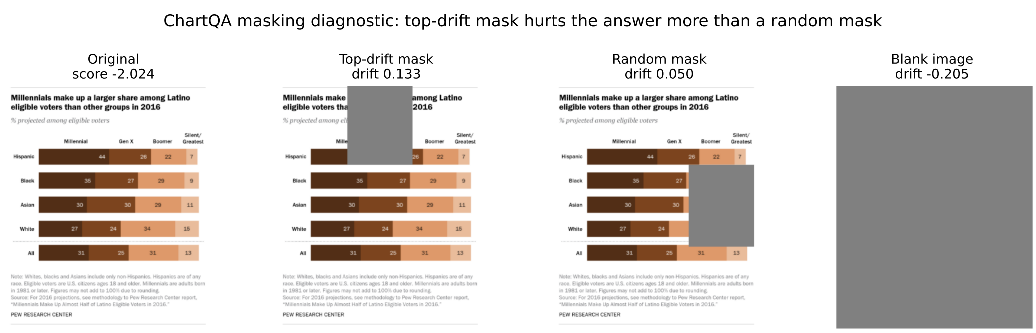

Figure 9: Example masking diagnostic for a ChartQA answer-only fallback run. The pipeline saves the original image, the top-drift mask, a random-region mask, and a blank-image control, together with the full-image and masked answer scores. This figure is used as an evidence-sensitivity example, not as proof that a generated rationale is faithful.

4.5. Efficiency and scale trade-offs

Table 11 and Figure 10 put Qwen2-VL-2B and Qwen2-VL-7B side by side on the same ScienceQA explanation-aware setting ($\alpha = 0.50$). Scaling up pays. The 7B run lifts accuracy from 0.765 to 0.885. ROUGE-L rises from 0.643 to 0.738, BLEU from 0.493 to 0.637. It is the single largest gain anywhere in the completed set. Both runs still fit under QLoRA. What the logs do not give us is a stable wall-clock or peak-memory figure, so this stands as a model-scale result, not a precise efficiency one.

Table 11: Qwen2-VL scale comparison on ScienceQA with the same explanation-aware objective ($\alpha = 0.50$).

Model	**MC accuracy**	**ROUGE-L**	**BLEU**
Qwen2-VL-2B	0.765	0.643	0.493
Qwen2-VL-7B	**0.885**	**0.738**	**0.637**

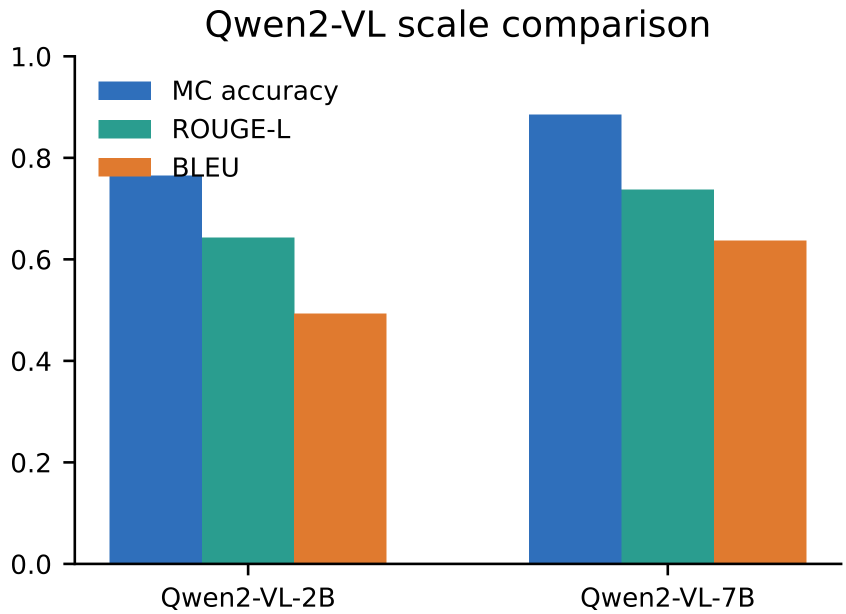

Figure 10: Scale comparison for the fixed ScienceQA explanation-aware setting. Moving from Qwen2-VL-2B to Qwen2-VL-7B improves both answer accuracy and rationale-overlap metrics.

4.6. Cross-domain transfer

Table 12 and Figure 11 lay out the train-domain against test-domain transfer matrix. Every column is scored in its target domain's own metric, so the only safe comparisons run down a column, not across a row. Where the controls exist, the matrix keeps answer-only, rationale-generative, and explanation-aware sources separate. Treat it as an evaluation probe rather than the official in-domain table; a diagonal cell can drift a little from the matching completed run when the transfer evaluator routes through a different prompt family. The probe matters because answer-only training often travels better than rationale-generative training. The ScienceQA answer-only adapter is the clearest case, reaching 0.805 on A-OKVQA, 0.645 on ChartQA, 0.703 on DocVQA, and 0.630 on VQAv2. Its rationale-generative and explanation-aware siblings clear the random baselines but fall off on ChartQA, DocVQA, and VQAv2. A-OKVQA sources split the honours: rationale-generative training wins DocVQA transfer in that group at 0.764, while answer-only wins VQAv2 at 0.700. DocVQA is strongest at home (0.910), and the document-trained adapter carries over to ChartQA too

(0.685). The pattern, in short: transfer depends on the target domain and the source objective together, not on domain alone.

Table 12: Cross-domain transfer matrix. Rows indicate the training domain and columns indicate the evaluation domain. Each column uses the target domain's headline metric, so values are best compared within a column.

Trained on	**ScienceQA**	**A-OKVQA**	**ChartQA**	**DocVQA**	**VQAv2**
ScienceQA answer-only	0.800	0.805	0.645	0.703	0.630
ScienceQA rationale CE	0.780	0.760	0.450	0.559	0.300
ScienceQA explanation-aware	0.740	0.765	0.460	0.491	0.410
A-OKVQA answer-only	0.690	0.825	0.635	0.677	0.700
A-OKVQA rationale CE	0.665	0.790	0.645	0.764	0.600
A-OKVQA explanation-aware	0.680	0.795	0.615	0.763	0.640
ChartQA answer-only	0.420	0.415	0.660	0.831	0.680
DocVQA answer-only	0.450	0.435	0.685	0.910	0.660
VQAv2 answer-only	0.440	0.575	0.660	0.716	0.730

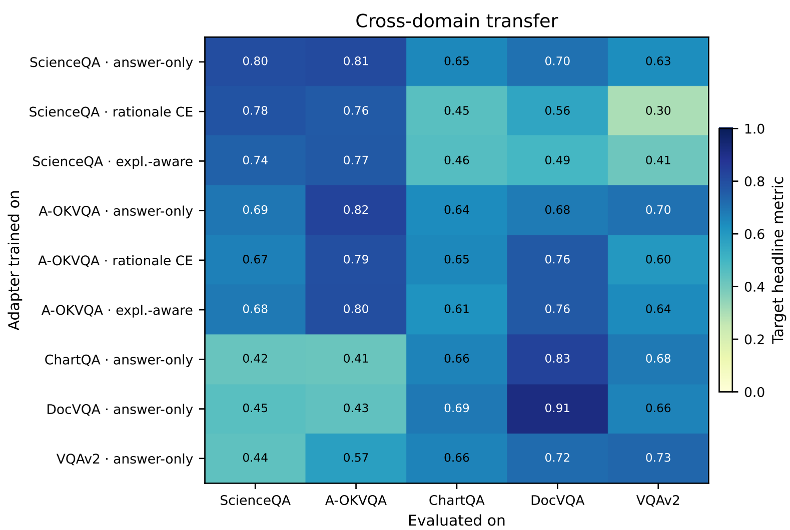

Figure 11: Cross-domain transfer heatmap. The strongest non-diagonal transfer is not always from the visually closest domain. Answer-only ScienceQA transfers strongly to A-OKVQA, while answer-only document and chart adapters transfer strongly to several open-ended targets.

5. Discussion

The results back the central idea, but only with caveats attached. They also slot neatly into the arc the literature review traced: VLM alignment has drifted from image–text representation matching toward generative instruction following, harder multimodal reasoning tests, and separate audits for hallucination, critique, and evidence use. Objective choice clearly matters; explanation supervision being automatically better than plain answer-only fine-tuning does not follow. It can help, yes, but only when it is balanced with care and held against a true answer-only control. On ScienceQA the best fixed explanation-aware run is the low-answer-weight one, and the length-aware control merely matches the rationale-generative control. The answer-span-only setting is plainly worse on both accuracy and rationale quality. A-OKVQA shows a small observed edge for explanation-aware training over the generative control, but the capped-run uncertainty is wide enough that we will not call it a firm win. And the fixed $\alpha = 0.50$ ScienceQA run is actually worse than both the rationale-generative and answer-only controls. Adding explanation tokens, by itself, does not make the model better. The loss balance is what does the work.

The BLIP-2 contrastive result earns attention precisely because it is so weakly positive on accuracy. We built the contrastive-enhanced arm specifically so the comparison could be run head-on. In the completed ScienceQA run it sits just 0.005 above the BLIP-2 generative control, well inside the approximate confidence intervals. Retrieval barely separates from the frozen and random controls either. The read is that the auxiliary image–answer InfoNCE term, at least in this small Q-Former-side setup, cannot forge a clearly stronger image–answer representation. For this experiment set, generative and explanation-aware supervision rest on firmer ground than contrastive enhancement.

The domain results are the clearest argument against trusting a single benchmark. ChartQA, DocVQA, VQAv2, A-OKVQA, and ScienceQA every one improves in the selected runs. Yet the reading stays mixed, because the domains are not all testing the same kind of supervision. ScienceQA and A-OKVQA carry rationale-bearing targets; ChartQA, DocVQA, and VQAv2 fall back to answer-only training, because no gold rationales are there to use. The strong DocVQA number, then, is capped extractive document answering, nothing more, not evidence of faithful document rationales and not a public-test DocVQA claim. Any stronger document-reasoning claim would first need document-specific rationale supervision, or a more causal OCR and layout faithfulness test.

The faithfulness numbers come with a warning attached. High masking drift proves the model is sensitive to visual change; it does not prove the explanation is faithful. DocVQA and ChartQA drift hard because they live on local visual evidence, and in this run both also improve under plain answer-only fallback. ScienceQA and A-OKVQA post better answer and rationale scores, yet their drift stays small and refuses to climb consistently with explanation-aware training. Better rationale text, in other words, is not the same animal as more faithful visual grounding. So the masking diagnostic earns a place beside BLEU and ROUGE-L, never a place instead of them.

Scale hands us the cleanest positive result of all. Holding the ScienceQA explanation-aware objective fixed, Qwen2-VL-7B beats the 2B model on both accuracy and rationale metrics. That does not retire the 2B experiments, since the small model is exactly what made the wide sweep affordable in the first place. What it does say is that some of the 2B weakness is probably about capacity, not objective.

Transfer is mixed in the same way. Answer-only ScienceQA jumps cleanly to A-OKVQA, while the rationale-generative and explanation-aware ScienceQA adapters lose ground on the open-ended targets. A-OKVQA repeats the shape: rationale-bearing objectives help on some targets, but answer-only is the stronger bet for VQAv2. The answer-only ChartQA and DocVQA adapters trade well with each other under the target metrics, a hint that structured, text-heavy images share something useful when it comes to adaptation. The conclusion is the same one again: transfer rides on the target domain and the source objective together.

5.1. Limitations

Compute is the first limitation, and the obvious one. A single-GPU budget keeps the study realistic, but it also caps the seeds, the model sizes, and the full-dataset runs we could attempt. Rationale availability is the second. ScienceQA and A-OKVQA ship gold explanations; DocVQA, ChartQA, and VQAv2 do not, so they cannot carry the same rationale supervision, and any teacher-generated rationale experiment on them must be reported apart from gold-rationale training. The metrics are the third. BLEU and ROUGE-L can wave off a perfectly valid explanation just because the wording differs, and while the masking metric gets closer to actual evidence use, it hinges on which regions are masked and turns noisy on dense documents and charts. The protocol is the fourth: capped train and evaluation subsets, one seed per full configuration. That is what makes the work feasible and auditable, but it also means small gaps are not statistically settled. And finally, the execution logs are good enough to validate the outputs, yet not good enough for precise wall-clock or peak-memory benchmarking. Taken together, these are the lines the paper will not cross in what it claims.

5.2. Future directions

The first thing future work should do is rerun the strongest configurations with more seeds and larger evaluation splits. After that, swap in newer open multimodal backbones while keeping the objective-isolation protocol untouched. A third push is better rationale annotation for chart and document datasets, the very places where faithful explanations would pay off most. The faithfulness test itself should grow more causal, editing document text, chart marks, or image regions in controlled ways rather than masking blindly. There is also a clean comparison waiting between supervised explanation-aware training and preference-based or reinforcement-learning methods that reward grounding directly. And before any of this ships, deployment work should track latency, memory, and calibration alongside accuracy and faithfulness.

6. Conclusion

This paper asked what the alignment objective alone does to vision–language reasoning under a fixed, resource-constrained fine-tuning budget. The comparison spanned zero-shot baselines, BLIP-2 generative and contrastive-enhanced alignment, and Qwen2-VL in answer-only, rationale-generative, and explanation-aware forms, across document, chart, science, commonsense, and natural-image benchmarks. Throughout, we measured answer accuracy, rationale quality, and evidence sensitivity together rather than one at a time. What holds up best in the completed runs is Qwen2-VL task adaptation: rationale supervision applied with care where gold rationales exist, answer-only fallback where they do not. BLIP-2 contrastive-enhanced training is here as a route we genuinely tested, but its gain over BLIP-2 generative training is small and its retrieval diagnostic is weak. All five datasets, ScienceQA, A-OKVQA, ChartQA, DocVQA, and VQAv2, improve under the selected runs. The faithfulness analysis lands on a distinction worth keeping: better explanation text and stronger visual evidence use are related, not identical. Explanation-aware training lifts rationale text in some settings, yet it does not reliably raise masking-based evidence dependence. Scale is the cleanest win, with Qwen2-VL-7B strengthening the ScienceQA explanation-aware result, while the transfer matrix shows alignment staying stubbornly target-domain dependent. The honest summary is a careful one: supervised rationales can help resource-constrained VLMs, but only when answer-only controls, domain shift, and the evidence test are weighed together.

Declarations

Author responsibility. The authors wrote, reviewed, and edited the manuscript and take full responsibility for the final content, results, and claims.

Use of generative AI. Generative AI tools were used only for language editing, code assistance, and consistency checks. All scientific decisions, experiment selection, result interpretation, and final manuscript approval remain the responsibility of the authors.

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Data, code, and artifacts. The experiments use public datasets cited in the paper. The code, configuration files, generated tables, figures, execution logs, and claim-to-evidence artifacts are maintained with the project repository and can be made available with the submitted supplementary material.

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